Contents

1 Step 10 — Population-Based Training and Evolutionary Game Theory: Experiment Report

Testbeds: four symmetric matrix games for the evolutionary-dynamics half (Prisoner's Dilemma, Hawk-Dove, Rock-Paper-Scissors, Stag Hunt); a synthetic pure-skill ladder; the PSRO best-response meta-game on Leduc Hold'em; and a small AlphaStar-style **PBT league** of neural PPO agents playing Leduc Hold'em. The Leduc engine, the exact best-response oracle, and NashConv/exploitability come from the Step 07 stack (reused wholesale); PSRO and the meta-Nash solver come from Step 09. Neural agents are trained with PyTorch and then extracted to *tabular* policies so the Step 07 **exact** best response can grade them.

PhD connection: the population-level layer of the thesis. Three hooks: **main exploiters as automated opponent modelers** (Contribution #1, the population lift of Step 07's Bayesian read); the **AlphaStar league as a population safe-exploitation mechanism whose guarantee is missing** (Contribution #2); and **EGTA / meta-Nash as the population evaluation methodology** (Contribution #3).

Scope of results: every number in this report is **measured from a real run** and read from the artifacts under `implementation/step10/implementation/results/{smoke,scale}_results.json` and `implementation/step10/exploration/figures/*.json`. Evolutionary results are bracketed by *analytic* references (ESS / Nash of each matrix game); league results are bracketed by Step 07's *exact* exploitability. Three Phase-4 predictions were **contradicted** by the scale run; per WORKFLOW §0.1 they are kept as stated and reconciled with what actually happened (§8).

How to read this report. Both parts follow the same arc: **what we test -> how -> results -> conclusion. Part I (§1-§3)** uses evolutionary game theory as a *diagnostic lens*: replicator dynamics on solvable matrix games, then the transitive/cyclic "spinning-top" decomposition. **Part II (§4-§7)** builds and evaluates the **PBT league**: training dynamics, EGTA/meta-Nash, diversity + Elo, and a head-to-head against PSRO / self-play / CFR-Nash. §8 reconciles the three contradicted predictions; §9-§11 cover trust, limitations, and directions; §12 lists reproduction commands.

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2 PART I — EVOLUTIONARY GAME THEORY AS A DIAGNOSTIC LENS

2.1 1. What this step is about

Steps 2-9 reasoned about equilibria of a *fixed* game. Step 10 moves up a level to **populations of strategies that change over time** and asks two questions a single-agent view cannot: *which strategies grow?* (replicator dynamics) and *does this game even have a "best" strategy, or only a wheel of counters?* (the transitive/cyclic decomposition). These two tools are the diagnostic that Part II's population-based training needs — because whether a self-training population converges or cycles is decided by the game's structure, not the learning rate. All Part I evolutionary numbers are deterministic (fixed seeds, exact dynamics), so "did it converge?" is unambiguous.

2.2 2. Experiment 1 — replicator dynamics on four matrix games

What we test. Do the replicator dynamics reproduce each game's *analytic* evolutionary outcome — convergence to a dominant strategy, to an interior ESS, to a basin-dependent pure ESS, or a non-convergent orbit? (Raw step validation, L530-533.)

How. Discrete-Euler replicator integration from a seeded interior start; "converged" means the last iterate stopped moving; "orbit radius" is the distance the trajectory holds from the uniform point. Analytic ESS/Nash is the reference. Data: `results/smoke_results.json` (`replicator` block), phase portraits in `exploration/figures/replicator_playground.json`.

Game	Analytic reference	Start x_0	Final x	Converged?	Orbit radius
Prisoner's Dilemma	Defect dominates; unique ESS $(0,1)$	$[0.666, 0.334]$	$[0.0, 1.0]$	yes	0.707
Hawk-Dove	interior ESS $p(\text{Hawk}) = V/C = 0.5$	$[0.548, 0.452]$	$[0.5, 0.5]$	yes	0.0
Rock-Paper-Scissors	Nash = uniform; **no ESS** (centre)	$[0.347, 0.385, 0.268]$	$[0.299, 0.288, 0.413]$	**no**	0.095
Stag Hunt ($x_0 = [0.8, 0.2]$)	two pure ESS (basin-dependent)	$[0.8, 0.2]$	$[1.0, 0.0]$	yes	0.707
Stag Hunt ($x_0 = [0.2, 0.8]$)	two pure ESS (basin-dependent)	$[0.2, 0.8]$	$[0.0, 1.0]$	yes	0.707

Results. All four match theory exactly. Prisoner's Dilemma collapses to all-Defect (the dominant strategy). Hawk-Dove converges to the interior mixed ESS at exactly 0.5 (orbit radius 0.0 — it *reaches* the rest point). Rock-Paper-Scissors **does not converge**: it orbits the uniform centre at a steady radius (0.095), because the interior fixed point is a centre, not an attractor. Stag Hunt lands on *different* pure ESS depending on the start — $[0.8, 0.2] \rightarrow$ all-Stag, $[0.2, 0.8] \rightarrow$ all-Hare — the two basins made concrete.

Conclusion. Replicator dynamics are the continuous idealization of population selection, and their rest points are exactly Nash/ESS. The one that *refuses to converge* — RPS — is the whole reason Part II exists: a cyclic game has no stable population, so a self-training population in it will spin rather than settle.

2.3 3. Experiment 2 — the spinning-top: transitive vs cyclic structure

What we test. Can we decompose a game's payoff matrix into a **transitive** (skill-ladder) component and a **cyclic** (rock-paper-scissors) component, and does the decomposition correctly identify

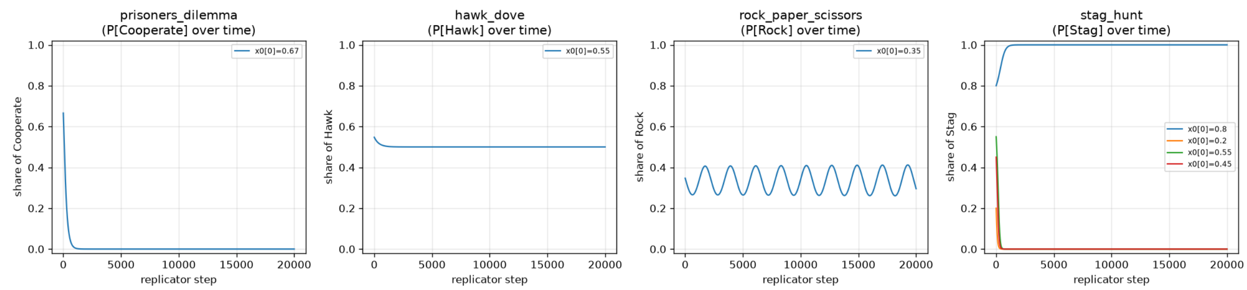

Figure 1: Replicator phase portraits on the four matrix games: Prisoner's Dilemma collapses to all-Defect, Hawk-Dove converges to the interior 0.5 ESS, Rock-Paper-Scissors orbits the centre without converging, and Stag Hunt splits into two basins (all-Stag or all-Hare) depending on the start. The non-converging RPS orbit is the visceral case for the diversity machinery in Part II.

pure cases and diagnose real populations? (Raw step validation, L532, L534.)

How. The combinatorial-**Hodge** (ratings-difference) decomposition on the antisymmetrized payoff matrix, reporting a transitive ratio and a cyclic ratio (orthogonal, so they combine in quadrature to 1). We also report the raw-step's SVD rank-1 transitive ratio for RPS to document its known failure. Data: `results/{smoke,scale}_results.json` (`spinning_top` block), `exploration/figures/game_landscape.json`.

Population	Transitive ratio (Hodge)	Cyclic ratio (Hodge)	Three-cycles	Note
Rock-Paper-Scissors	0.0	1.0	1	SVD rank-1 wrongly gives 0.707
Pure skill ladder	1.0	0.0	0	monotone strength
PSRO-Leduc meta-game (smoke, 8 rounds)	0.457	0.890	—	mostly cyclic
PSRO-Leduc meta-game (scale, 20 rounds)	0.411	0.912	27	mostly cyclic

Population	Transitive ratio (Hodge)	Cyclic ratio (Hodge)	Three-cycles	Note
League snapshot meta-game (smoke / scale)	0.981 / 0.937	—	—	mostly transitive

Results. The Hodge decomposition nails the pure cases: RPS is 0.0 transitive / 1.0 cyclic (a single three-cycle), the skill ladder is 1.0/0.0. The **SVD rank-1** method the raw step suggested instead reports RPS as ≈ 0.707 transitive — wrong, because a rank-1 skew-symmetric approximation cannot represent a pure cycle; the Hodge method is the one used for the diagnosis. The two *real* populations diverge sharply: the **PSRO best-response meta-game on Leduc is mostly cyclic** (≈ 0.41-0.46 transitive, 27 three-cycles), while the **league's snapshot meta-game is mostly transitive** (≈ 0.94-0.98).

Conclusion. The transitive/cyclic ratio is a *pre-training diagnostic*. Two populations drawn from the same game (Leduc) have opposite structure depending on how they are built (best responses vs training snapshots — see §8.2). This is exactly the tool that predicts whether naive self-play/PBT will converge or cycle, and it is the diagnostic the thesis carries into later steps.

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3 PART II — THE PBT LEAGUE

3.1 4. What the league is and how it is graded

The second half builds a small **AlphaStar-style league** on Leduc Hold'em with three agent types — **main** agents (the product), **main exploiters** (hunt weaknesses in the current mains), and **league exploiters** (hunt weaknesses anywhere in the frozen history) — plus periodic **freezing** of snapshots into a museum and **PFSP** matchmaking (sample harder opponents more often). Agents are neural PPO networks on a Leduc info-state encoding; PBT copies the top agents (exploit) and perturbs their learning rate / entropy (explore). Crucially, each network is extracted to a **tabular** policy so Step 07's **exact** best response measures its exploitability — the same NashConv yardstick used since Step 2. Two configs: **smoke** (7 live agents, 15 epochs) and **scale** (8 live agents, 120 epochs, 48 frozen snapshots).

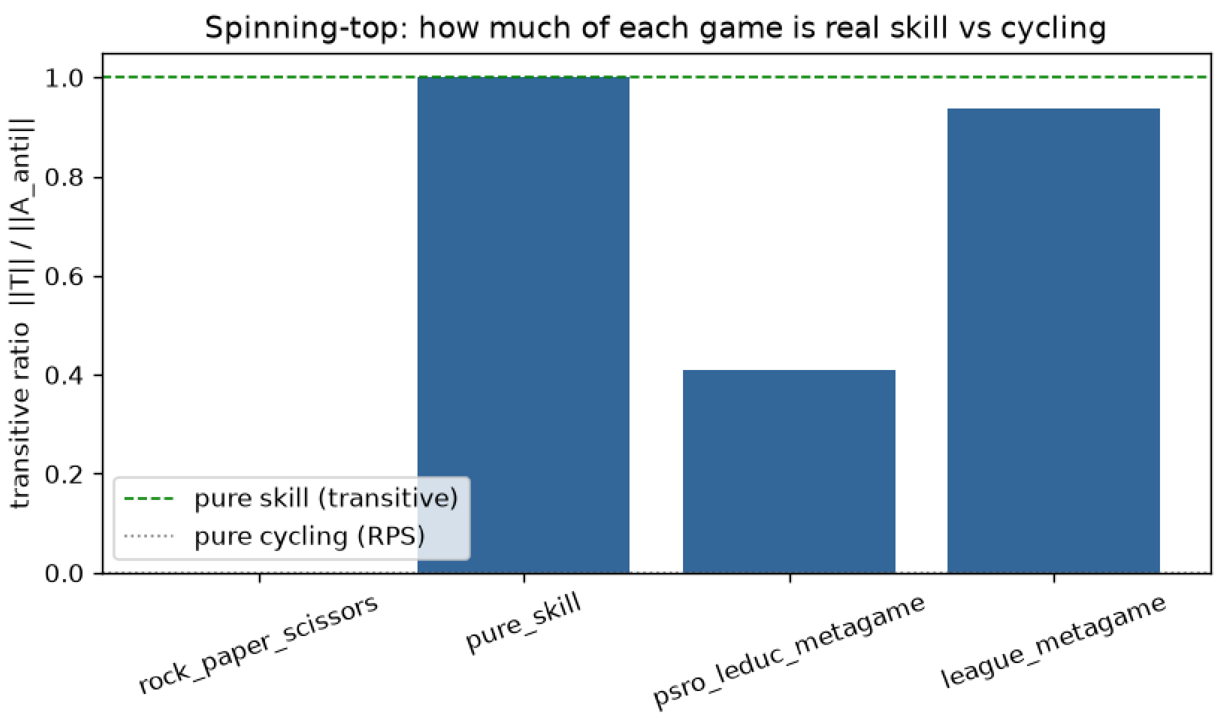

Figure 2: Transitive-ratio bar chart: Rock-Paper-Scissors (0.0, purely cyclic), a pure skill ladder (1.0, purely transitive), the PSRO-Leduc best-response meta-game (~0.41-0.46, mostly cyclic with 27 three-cycles), and the league snapshot meta-game (~0.94-0.98, mostly transitive). Which population you decompose decides whether Leduc looks like a wheel or a ladder.

3.2 5. Experiment 3 — league training dynamics

What we test. Does the league's exploitability fall as training proceeds, and does the meta-Nash of the population track that improvement? (Raw step validation, L535, L537.)

How. Every epoch, record the minimum exploitability across the main agents and the exploitability of the current meta-Nash mixture; also record the league snapshot meta-game's transitive ratio and final Elo. Data: `results/{smoke,scale}_results.json` (`league` block).

Metric	Smoke (15 epochs)	Scale (120 epochs)
min-main exploitability	4.67 → 3.04 (monotone, ends at min)	4.73 → **min** ≈ 1.21 (ep ~64) → 2.05 (ep 119)
meta-Nash exploitability	4.73 → 3.04	5.01 → min ≈ 1.32 (plateau 1.60) → 2.96 (ep 119)
league meta-game transitive ratio	0.981	0.937
final Elo (live agents)	1176-1211	1198-1210

Results. Smoke shows a clean monotone decline and ends at its minimum (3.04) — but 15 epochs is too short to show what happens next. Scale reveals it: exploitability falls steeply to a minimum near epoch 60-65 (**min-main** ≈ 1.21, meta-Nash bottoming ≈ 1.32 then holding a ≈ 1.60 plateau), and then **regresses** back up to ≈ 2.05 (min-main) and ≈ 2.96 (meta-Nash) by epoch 119. Elo spread is compressed (all live agents within ≈ 15 points), consistent with a behaviorally-similar population.

Conclusion. The league *does* drive strong early improvement — but improvement is **not monotone at scale**: under sustained exploiter pressure the live main agents chase their exploiters and lose ground on absolute exploitability late in training. The best agents are captured as *frozen snapshots* mid-run. This non-monotonicity is the honest headline of the scale run and is reconciled in §8.3.

3.3 6. Experiment 4 — EGTA, meta-Nash, and diversity

What we test. (a) Is the meta-Nash of the final population less exploitable than any individual agent (the raw-step expectation, L536)? (b) How diverse is the population (effective size, behavioral clustering, exploit coverage — L538)?

How. Build the empirical symmetric payoff matrix over all agents (live + frozen), solve the meta-Nash mixture, collapse it to a single behavioral policy, and measure its exact

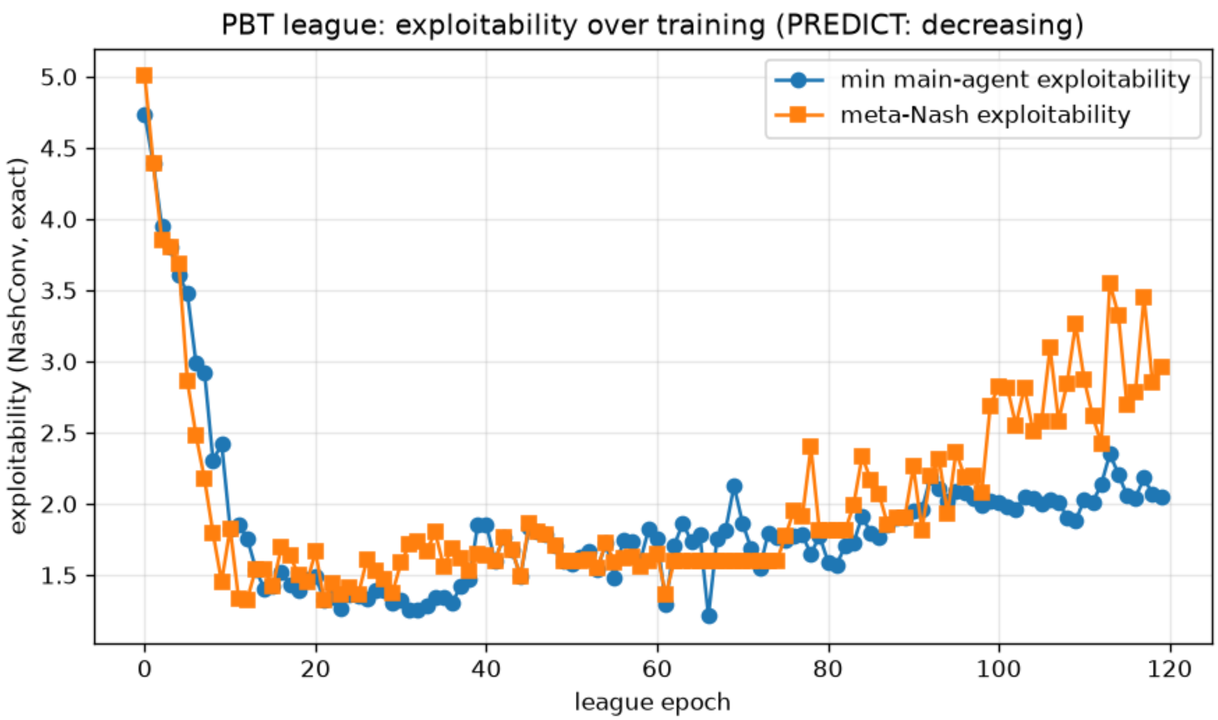

Figure 3: League exploitability over training (scale, 120 epochs): min-main and meta-Nash exploitability both fall steeply to a minimum near epoch 60 (min-main ~1.21, meta-Nash ~1.32) and then regress upward (~2.05 / ~2.96) by epoch 119. A running league is not a monotonically improving one; the best agents are frozen snapshots from mid-run.

exploitability against the best individual's. Diversity: effective population (participation ratio of meta-Nash weights), single-linkage behavioral clustering, and exploit coverage. Data: `results/{smoke,scale}_results.json` (`league.egta`, `league.diversity`). Toy intuition: `exploration/figures/mini_pbt.json`.

(a) Meta-Nash vs best individual:

Config	meta-Nash exploitability	best individual	meta-Nash $\leq$ best?
Smoke	2.665	2.665	**yes** (all weight on the best agent)
Scale	3.418	1.305	**no** (weight spread `0.645`+tail)

(b) Diversity:

Metric	Smoke	Scale
agents / active (participation)	16 / 1 (1.00)	56 / 3 (1.92)
behavioral clusters (thr 0.30)	1 (max pairwise 0.255)	1 (max pairwise 0.484)
exploit coverage	0.625	0.571

Results. (a) In smoke the meta-Nash puts all weight on the single best agent, so meta-Nash = best = 2.665 (the prediction holds trivially). At scale the meta-Nash spreads weight and its collapsed mixture is *more* exploitable (3.418) than the best single member (1.305) — the prediction **fails** (§8.1). (b) The population is only weakly diverse: participation ratio rises from 1 to 1.9, but behavioral clustering collapses everything into a **single cluster** at both scales (even though the scale population's max pairwise distance 0.484 exceeds the 0.30 threshold — single-linkage merges the chain). The mini-PBT toy makes the mechanism vivid: on transitive Prisoner's Dilemma diversity collapses to 0; on cyclic RPS it churns forever (0.07-0.29).

Conclusion. EGTA gives a population-level exploitability, but two lessons emerge: mixing behavioral policies via the meta-Nash does **not** guarantee a less-exploitable result (§8.1), and PBT diversity here is *weight-level*, not *behavior-level* — the agents are near-identical, so the "population" is thinner than its size suggests.

3.4 7. Experiment 5 — league vs PSRO vs self-play vs CFR-Nash

What we test. How does the league compare to the alternatives — exact PSRO, plain self-play, and CFR-Nash — on Leduc exploitability? (Raw step validation, L539.)

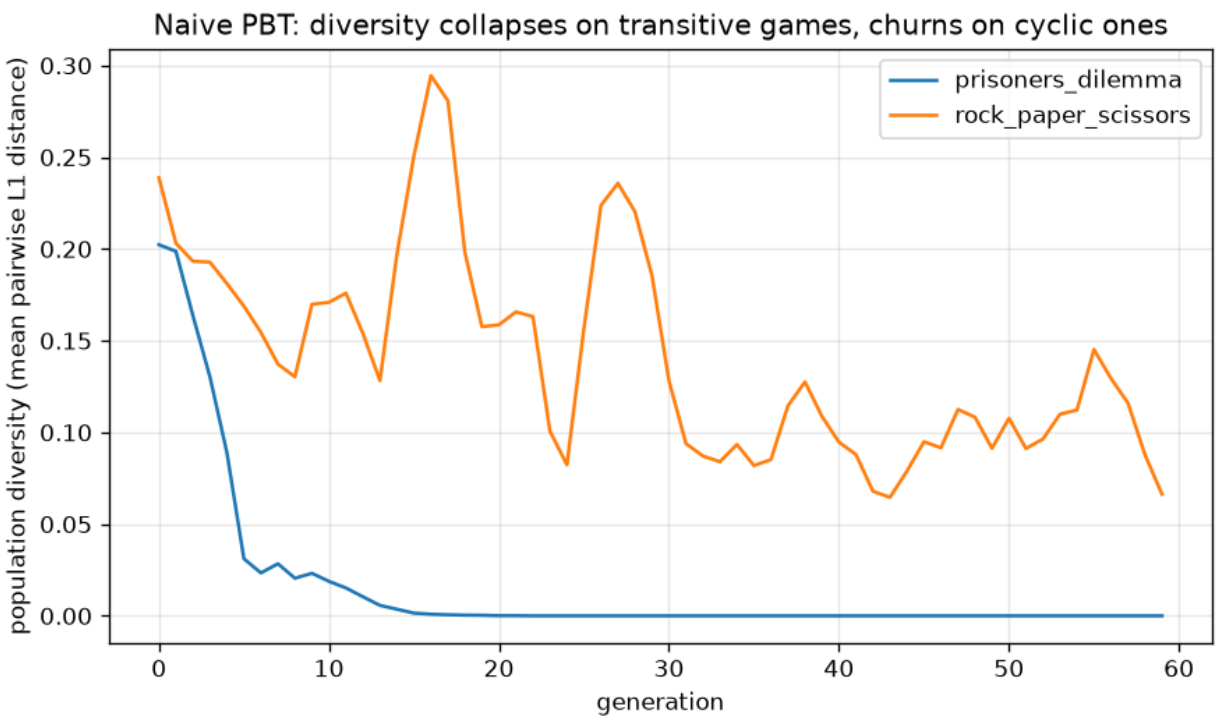

Figure 4: Mini-PBT diversity over generations: on the transitive Prisoner's Dilemma the population's strategy diversity collapses to zero (everyone converges to Defect), while on cyclic Rock-Paper-Scissors it churns indefinitely (0.07-0.29) as the population chases the wheel of counters. Game structure, not population size, sets whether diversity survives.

How. Run each method to its config's budget and read the final exploitability. The league entry is its **best individual** (the strongest frozen snapshot); PSRO uses the exact BR oracle; CFR-Nash is the approximate-Nash floor. Data: `results/{smoke,scale}_results.json` (`league`, `baselines`).

Method	Smoke	Scale	Note
CFR-Nash (floor)	0.033 (2k iters)	0.0099 (20k iters)	approximate Nash, the target floor
League — best individual	2.665	1.305	strongest frozen snapshot
PSRO (exact BR)	3.037 (12 rounds)	2.163 (20 rounds)	Step 09's slow-Leduc wall
Self-play	3.140	3.683	plain best-response-to-latest
League — meta-Nash mixture	2.665	3.418	see §8.1

Results. At scale the league's **best individual** (1.305) is the strongest of the learned methods — better than exact PSRO (2.163) and self-play (3.683) — though all remain far above the CFR-Nash floor (0.0099). The league's **meta-Nash mixture** (3.418), by contrast, is the *weakest*, worse even than self-play (§8.1). Self-play actually *worsens* from smoke to scale, consistent with last-iterate non-convergence.

Conclusion. The league is a good producer of strong *individuals* — its best snapshot beats PSRO and self-play — but the *mixture* is not, and none of the learned methods approach the exact-CFR floor on Leduc. The right thing to ship from a league is a *selected member*, not the meta-Nash mixture (§8.1).

3.5 8. Prediction <-> reality reconciliation

Per WORKFLOW §0.1, contradicted Phase-4 predictions are kept and reconciled, not silently edited. Three gaps.

1. **Meta-Nash of the league $\leq$ best individual (§6a).** *Predicted* (raw step Exit Checklist, L536): the meta-Nash is less exploitable than any individual agent. *Measured:* true in smoke (2.665 = 2.665, all weight on the best agent), **false at scale** (3.418 > 1.305). *Reconciliation:* not a mixing bug — the identical `mixture_behavioral_policy` code path gave meta = best in smoke. The meta-Nash minimizes **meta-game regret** (win against the population), which is a *different objective* from minimizing **full-game exploitability**; and a realization-weighted

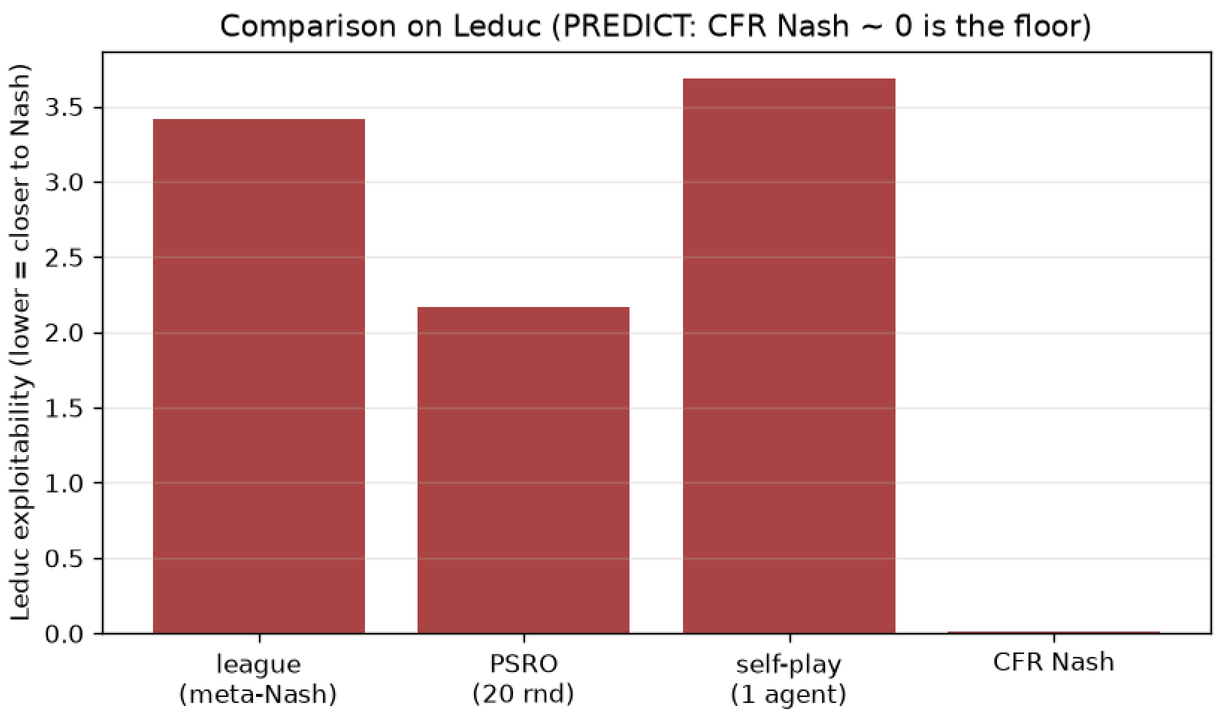

Figure 5: Final Leduc exploitability by method (scale): CFR-Nash floor ~0.01; the league's best individual (1.31) beats PSRO (2.16) and self-play (3.68); the league's meta-Nash mixture (3.42) is the weakest learned result. Ship a selected member, not the mixture.

mixture of behavioral policies can be *more* exploitable than its components, because the mixture introduces info-set "tells" a best responder punishes. The lesson inverts and sharpens: population evaluation must measure the collapsed mixture's exploitability directly and, in practice, ship a selected member rather than the mixture.

2. **Leduc's meta-game is transitive (§3).** *Predicted:* the intuition doc framed poker as a skill ladder. *Measured:* the PSRO **best-response** meta-game is mostly cyclic ($\approx$ 0.41-0.46 transitive, 27 three-cycles), while the league **snapshot** meta-game is mostly transitive ($\approx$ 0.94-0.98). *Reconciliation:* not a bug — a population of best responses cycles (A beats the mixture, B beats A, a later BR beats B; Balduzzi's spinning-top), whereas a population of training-trajectory snapshots forms a ladder because later snapshots are mostly stronger. The structure is a property of the *population*, not just the game; both measurements are correct and instructive.

3. **League exploitability decreases monotonically (§5).** *Predicted:* a monotone decline. *Measured:* smoke (15 epochs) declines cleanly and ends at its minimum (3.04); scale (120 epochs) falls to a minimum near epoch 60 (min-main $\approx$ 1.21, meta-Nash $\approx$ 1.32/plateau 1.60) and then **regresses** to $\approx$ 2.05 / $\approx$ 2.96 by epoch 119. *Reconciliation:* an honest, likely-real training instability, only visible once training is long enough — under sustained exploiter pressure the live main agents chase their exploiters and lose absolute-exploitability ground (churn / partial forgetting). The best agents are the frozen snapshots from mid-run, which is exactly why the best-individual metric (§7) is strong while the late live agents are not. Documented, not fixed; the natural remedy is best-snapshot retention / population regularization (§11). This echoes Step 09's methodological rule: **scale reveals what smoke hides.**

3.6 9. Trustworthiness and sample adequacy

- **Evolutionary results are bounded by analytic references.** Each replicator outcome is checked against the game's analytic ESS/Nash; the spinning-top ratios are checked against the pure cases (RPS = 0.0, skill ladder = 1.0). These are deterministic and exactly reproducible.
- **League exploitability is Step 07's *exact* NashConv**, not a simulation: every neural policy is extracted to a tabular policy and scored by exact best response, so "how exploitable is this agent/mixture?" is ground-truthed rather than estimated.
- **Neural training is seed- and library-sensitive.** The league is one PBT run per config; the *qualitative* findings (early improvement; best-individual < PSRO < self-play; late regression at scale; meta-Nash > best member at scale) are the trustworthy claims, not the third-decimal magnitudes.
- **Diversity metrics are threshold-sensitive.** The single-cluster result depends on the 0.30

single-linkage threshold; it should be read as "behaviorally similar," not "provably one strategy."

3.7 10. Limitations (ranked by how much they affect the conclusions)

1. **Late-training regression at scale (§5, §8.3)** — the league does not monotonically improve; until best-snapshot retention/regularization is added and re-run, the "league improves" claim holds only for the *first half* of training and for the *frozen best*, not the live agents.
2. **Meta-Nash mixture more exploitable than the best member (§6a, §8.1)** — means population evaluation must not report the mixture as the league's strength; the selected member is the right artifact.
3. **Thin behavioral diversity (§6b)** — participation ratio 1.9 and a single behavioral cluster mean the "population" is near-degenerate; the diversity benefits AlphaStar reports at ~600 agents do not appear at this scale.
4. **Single-run neural results (§9)** — direction-robust, magnitude-uncertain.
5. **Toy scale throughout** — Leduc and small matrix games; nothing here should be extrapolated to deep-RL leagues at AlphaStar scale.

3.8 11. Conclusions and research directions

Conclusions. Part I delivers the diagnostic: replicator dynamics reproduce every analytic ESS/Nash (including RPS's non-converging orbit), and the Hodge spinning-top decomposition cleanly separates skill from cycles — revealing that the *same* game (Leduc) is cyclic as a best-response population but transitive as a snapshot population. Part II builds the league and grades it with exact exploitability: it produces strong *individuals* (best snapshot 1.305, beating PSRO 2.163 and self-play 3.683), but three predictions broke honestly — the meta-Nash mixture is *more* exploitable than its best member at scale, the Leduc best-response meta-game is *cyclic* not transitive, and league exploitability *regresses* late in long training. The population machinery works, but carries no monotonicity or safety guarantee — which is precisely the thesis gap.

Research directions (each tied to a measured effect):

- *Best-snapshot retention / population regularization* — directly targets the late regression (§8.3); re-run to see whether the guarantee gap is a training or a design issue.
- *Report the selected member, or a diversity-regularized meta-solver* — the meta-Nash-mixing failure (§8.1) motivates either shipping a best-response-robust member or changing the meta-solver objective.

- *Carry the transitive/cyclic diagnostic to Step 11's FFA games* — test the "large cyclic component" prediction directly, where naive PBT is expected to cycle.
- *Formalize population safety without a minimax anchor (Contribution #2)* — the exploiter mechanism is heuristic; §8.1/§8.3 show it does not guarantee a non-exploitable population.

3.9 12. Reproduction

From `implementation/step10/implementation/` with the project `.venv` active:

```
# validation harness (PASS/FAIL/SKIP against the raw-step targets)
python validate.py

# full comparison runs -> results/{smoke,scale}_results.json
python tournament.py --config smoke
python tournament.py --config scale

# plots from a results JSON -> plots/*.png  (needs matplotlib)
python plotting.py --config scale
```

From `implementation/step10/exploration/` (Part I figures + JSON):

```
python replicator_playground.py     # replicator phase portraits + JSON
python game_landscape.py            # transitive/cyclic ratios (RPS / skill / PSRO-Leduc)
python mini_pbt.py                  # diversity collapse vs churn
python psro_population_peek.py      # PSRO population dynamics on Leduc
```

Seeds are fixed in `config.py` (implementation) and each exploration script's config. The evolutionary and spinning-top results are exactly reproducible; the neural league results are direction-stable across seeds. All numbers in this report were read from `results/{smoke,scale}_results.json` and `exploration/figures/*.json`.